Bharath Kumar

Data Science & Gen Ai Engineer

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Location: Bengaluru

PROFILE SUMMARY

AI-ML Specialist with 12+ years of industry experience across Manufacturing, AI & Retail, Healthcare, and Supply Chain, leveraging deep expertise in Python, Flask, and AI/ML technologies. Over 2.5 years of hands-on experience in deep learning, Generative AI, and prompt engineering, delivering innovative end-to-end solutions. Skilled in managing the full ML lifecycle using tools like AWS Cloud, Google Cloud AI Platform, and Vertex AI.

AWARDS & TECH EXPOSURE

- I am Certified Scrum Master.

EXPERIENCE

Mphasis, Bengaluru
— Project Delivery Lead — July 2023 – PRESENT

- Led a cross-functional team of software engineers in the design, development, and deployment of scalable solutions, while mentoring junior engineers and fostering a collaborative environment focused on best practices for coding, testing, and system architecture.
- Leveraged expertise in Python, Data Science, Machine Learning, and Generative AI on Google Cloud Platform (GCP) to design, develop, and deploy scalable AI-driven solutions, optimizing performance and driving innovation and optimizing complex solutions across various projects.
- Developed and deployed scalable Python applications using Flask and Vertex AI on Google Cloud, integrating machine learning models to deliver intelligent, data-driven solutions.

Nine leaps Technologies, Bengaluru
— Principle Engineer — December 2019 – November 2021

Vitech Systems Asis Pvt Ltd, Hyderabad
— Software Developer— November 2017 – November 2019

Cognizant Technologies, Bengaluru
— Senior Infra Developer — June 2013 - September 2019

SKILLS

Technical Skills:

- Gen AI,
- Prompt Engineering
- Python, Flask
- Machine Learning
- Deep Learning
- Auto ML Vision, Video Intelligence, Visual Inspection
- TensorFlow, OpenCV
- Google Cloud Platform
  - Cloud Storage & Monitoring
  - Pub/Sub, Dataflow
  - VPC Network & IAM etc.,
- Doc AI, OCR
- Vertex AI, Big Query
- ThoughtSpot, Looker
- Aws Beddrock

DEPLOYMENT ENGINE:

- Compute Engine
- App Engine, Cloud run
- Google Kubernetes Engine
- API Gateway, Nginx
- Cloud Functions, Docker
- Auto ML Vision, Video
- Amazon Web Servies

- Industry Experience across Manufacturing, AI & Retail, Healthcare, and Supply Chain; proficient in Source Code Management tools like GitHub, Inner Source, SVN, Bitbucket, AWS and G-Cloud Source Repository;

PROJECTS

AI, Generative AI & Retail Industry

Project Name — Fiserv

- Managed business analysis for OTIF, CFR, on-time percentage, RCA, and inventory forecasting.
- Leveraged Generative AI (Gemini-pro-1.0, Text Bison, Code Bison) and Google-Certified Architecture to optimize processes.
- Implemented natural language search, top 3 recommendations, and data graph representation for user queries.
- Built AI bot using Amazon Lex (NLU) + Bedrock.
- Fast API backend with Lambda, DynamoDB.
- Used MCP + LLM routing.
- worked on multiple agent creation
- Features: Top 3 Recommended Questions, Feedback Sharing, Amazon Connect with voice assistant & multilingual conversation etc.,
- Improved response time by 40%.

Project Name — **Supply Chain Management (Agentic AI)**

- Developed a solution to extract **custom entities** from **contract documents** using **Agentic-AI models**.
- Enabled **real-time prompt engineering** and **graph-based responses** for improved decision-making.
- Streamlined the management of large **contract datasets** through intelligent **entity extraction**.
- Created an agentic AI system using LangChain, LangGraph, and LLaMA 3 8B, coordinating between suppliers, distributors, and warehouse systems.
- Built ReactJS + FastAPI dashboards for real-time analytics and predictive restocking.
- Integrated Groq API for faster inference and LangSmith for tracing and optimization.

Project Name — kore.ai,

- Led a cross-functional team in designing and deploying intelligent chatbots on the Kore.ai platform for multiple clients, consistently delivering high-quality solutions within tight deadlines. Demonstrated strong leadership by coordinating task ownership, mentoring junior developers, and ensuring alignment with client expectations.
- Maintained a flawless delivery record, never missing a deadline, while contributing hands-on to bot architecture, NLP configuration, and integration workflows.
- Successfully delivered 32 chatbots within an 8-month period for clients across diverse industries and domains.

Project Name — AI Driven Solutions (accelerators)

Safety Surveillance: Automated safety tracking in the construction industry using computer vision to monitor safety gear (hats, vests, shoes). Provided real-time alerts for non-compliance.

AIVI Utility Pole Inspection: Developed a solution to inspect utility pole assets using **Auto-ML Vision** and drones, identifying damage to transformers, arrestors, and insulators.

AI-Connected Worker: Tracked real-time worker movements using **video analytics** to ensure safety compliance, integrating **OpenCV**, **Auto-ML Video Intelligence**, and **IoT Core** for monitoring.

Shelf Stock Management: Developed a **web-based tool** for **retail inventory management**, using **IP cameras**, **SSD-Mobilenet-V3**, and **Python Flask** for real-time stock updates and product monitoring

Supply Chain & Manufacturing Industry

Manufacturing Suite: Developed an end-to-end solution for transferring data from SCADA, PLCs, SAP, and shop floors to the cloud using **Manufacturing Data Engine** and **Manufacturing Connect**. Utilized automated pipelines with **Pub/Sub**, **Big Query**, **SQL**, **Data Flow**, **Looker**, and **Vertex AI** to ensure seamless data flow. Enabled efficient end-to-end pipeline data transfer and analytics from the shop floor to the cloud.

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Education : Bachelor of Technology in Electrical and Electronics Engineering. JNTU-Anantapur (2008-2012)